

Tech School Lab Equipment Lists with Pricing (INR)

Innovative Tech School Setup for Grades 6+ | From Basic to Advanced Learning

ROBOTICS & AI LAB

BASIC EQUIPMENT (Starter Level)

Item	Quantity	Unit Price (INR)	Total Price (INR)	Specifications
Core Robotics Kits				
Quarky Ultimate AI-ML Robotics Kits	15 sets	8,500	1,27,500	Block coding + Python, AI/ML projects
Arduino Uno R3 Starter Kits	10 sets	2,500	25,000	Basic electronics + programming
Raspberry Pi 4 (4GB) Kits	8 sets	8,500	68,000	Complete starter kits with accessories
Basic Sensor Modules Kit (37-in-1)	10 sets	1,500	15,000	Ultrasonic, temperature, light, etc.
Programming Hardware				
Budget Laptops/Tablets for Programming	15 units	25,000	3,75,000	i3 processor, 4GB RAM, Windows
Assembly & Tools				
Breadboards (Half-size)	30 pieces	150	4,500	Solderless prototyping
Jumper Wire Sets	20 sets	200	4,000	Male-male, female-female
Basic Tool Kit (screwdrivers, pliers)	5 sets	800	4,000	Essential hand tools

Digital Multimeters	8 pieces	1,200	9,600	Basic measuring instruments
Storage & Organization				
Component Storage Boxes	10 pieces	500	5,000	Organized component storage
Mobile Trolleys for Equipment	3 pieces	4,000	12,000	Transport equipment between classes

BASIC EQUIPMENT SUBTOTAL: ₹6,49,600

ADVANCED EQUIPMENT (Pro Level)

Item	Quantity	Unit Price (INR)	Total Price (INR)	Specifications
Advanced Robotics				
Humanoid Robot Kits (NAO/Pepper style)	2 units	1,50,000	3,00,000	Professional humanoid platforms
Quadruped Robot Dog Kits	3 units	45,000	1,35,000	Advanced locomotion & AI
Robotic Arm Kits (6-DOF)	4 units	35,000	1,40,000	Industrial-grade manipulation
Mars Rover Building Kits	6 units	12,000	72,000	Space exploration simulation
AI & Vision				
AI Camera Modules (OpenCV ready)	10 units	8,000	80,000	Computer vision projects
VR Headsets for AR/VR Development	4 units	30,000	1,20,000	Meta Quest or equivalent
AI Edge Computing Modules	6 units	15,000	90,000	Jetson Nano equivalent
Advanced Programming				

High-end Programming Workstations	8 units	55,000	4,40,000	i7, 16GB RAM, dedicated graphics
Machine Learning GPU Cards	4 units	45,000	1,80,000	CUDA-enabled for ML training
Advanced Manufacturing				
3D Printer (Professional Grade)	2 units	1,50,000	3,00,000	Large build volume, precision
Laser Engraver/Cutter	1 unit	2,00,000	2,00,000	CO2 laser for prototyping
CNC Mini Mill	1 unit	1,80,000	1,80,000	Precision machining
Professional Tools				
Oscilloscopes (Digital)	4 units	25,000	1,00,000	Signal analysis
Logic Analyzers	3 units	18,000	54,000	Digital signal debugging
Soldering Stations (Professional)	6 units	8,000	48,000	Temperature controlled

ADVANCED EQUIPMENT SUBTOTAL: ₹23,39,000

FURNITURE & INFRASTRUCTURE

Item	Quantity	Unit Price (INR)	Total Price (INR)	Specifications
Furniture				
Modular Lab Tables (Adjustable Height)	8 units	12,000	96,000	6-student capacity each
Ergonomic Lab Stools	30 pieces	3,500	1,05,000	Adjustable, swivel base

Teacher's Demo Table with Storage	1 unit	18,000	18,000	Mobile, equipment storage
Equipment Storage Cabinets	6 units	15,000	90,000	Lockable, ventilated
Interactive Whiteboard/Smartboard	1 unit	80,000	80,000	75-inch, touch-enabled
Power & Infrastructure				
UPS Power Backup System (5KVA)	2 units	45,000	90,000	Lab-wide power protection
Network Switch (24-port managed)	2 units	12,000	24,000	Gigabit connectivity
WiFi Access Points (Enterprise grade)	2 units	8,000	16,000	High-density support
Power Distribution Units	8 units	3,500	28,000	Safe power distribution

FURNITURE & INFRASTRUCTURE SUBTOTAL: ₹5,47,000

ROBOTICS & AI LAB TOTAL:

- **Basic Equipment:** ₹6,49,600
- **Advanced Equipment:** ₹23,39,000
- **Furniture & Infrastructure:** ₹5,47,000
- **GRAND TOTAL:** ₹35,35,600

CODING & PROGRAMMING LAB

BASIC EQUIPMENT (Starter Level)

Item	Quantity	Unit Price (INR)	Total Price (INR)	Specifications
Computing Hardware				
Student Computers (Desktop)	25 units	35,000	8,75,000	i5, 8GB RAM, 256GB SSD
Teacher Workstation	1 unit	65,000	65,000	i7, 16GB RAM, dual monitors
Backup Laptops	5 units	28,000	1,40,000	For flexibility/backup
Development Tools				
Programming Keyboards (Mechanical)	30 units	2,500	75,000	Durable, tactile feedback
Ergonomic Mouse Sets	30 units	800	24,000	Optical, wireless
Dual Monitor Setup (Secondary)	25 sets	8,000	2,00,000	21-inch, IPS panels
Software Licensing				
Educational Software Suite	1 license	50,000	50,000	IDEs, compilers, design tools
Classroom Management Software	1 license	25,000	25,000	Monitor/control student screens
Basic Electronics for IoT				
Microcontroller Learning Kits	15 sets	3,500	52,500	Arduino/ESP32 based
Basic Sensor Kits for IoT Projects	10 sets	2,000	20,000	Temperature, motion, light

BASIC EQUIPMENT SUBTOTAL: ₹15,26,500

ADVANCED EQUIPMENT (Pro Level)

Item	Quantity	Unit Price (INR)	Total Price (INR)	Specifications
High-Performance Computing				
Gaming/Development Workstations	12 units	85,000	10,20,000	i7/Ryzen 7, 32GB RAM, RTX graphics
Server Rack for Cloud Computing	1 unit	2,50,000	2,50,000	Multi-core, high RAM
Advanced Development				
VR Development Setup	4 sets	75,000	3,00,000	VR headset + development PC
Mobile App Development Stations	8 sets	45,000	3,60,000	iOS/Android development
AI/ML Development				
GPU Computing Cluster	1 system	4,00,000	4,00,000	Multi-GPU for ML training
AI Development Workstations	6 units	1,20,000	7,20,000	High-end GPU, large RAM
Professional Software				
Professional IDE Licenses	1 package	1,50,000	1,50,000	Visual Studio, JetBrains suite
Game Engine Licenses	1 package	75,000	75,000	Unity Pro, Unreal Engine
Advanced Prototyping				
IoT Development Lab	1 setup	1,25,000	1,25,000	ESP32, sensors, cloud connectivity
Drone Programming Kits	6 units	25,000	1,50,000	Programmable drones

ADVANCED EQUIPMENT SUBTOTAL: ₹33,50,000

FURNITURE & INFRASTRUCTURE

Item	Quantity	Unit Price (INR)	Total Price (INR)	Specifications
Furniture				
Computer Lab Desks (Ergonomic)	13 units	8,000	1,04,000	2-student capacity each
Ergonomic Programming Chairs	30 pieces	4,500	1,35,000	Lumbar support, adjustable
Instructor Station	1 unit	25,000	25,000	Raised platform, storage
Equipment Racks	4 units	12,000	48,000	Server/networking equipment
Display & Interaction				
Interactive Projector System	1 unit	1,20,000	1,20,000	Ultra-short throw, 4K
Student Response System	30 units	1,500	45,000	Interactive voting/feedback
Networking & Power				
Enterprise Network Infrastructure	1 system	80,000	80,000	Managed switches, firewall
UPS Systems (3KVA)	3 units	35,000	1,05,000	Distributed power backup
Cable Management System	1 system	25,000	25,000	Professional cable routing

FURNITURE & INFRASTRUCTURE SUBTOTAL: ₹5,87,000

CODING & PROGRAMMING LAB TOTAL:

- **Basic Equipment:** ₹15,26,500
- **Advanced Equipment:** ₹33,50,000
- **Furniture & Infrastructure:** ₹5,87,000
- **GRAND TOTAL:** ₹54,63,500

 MIXED TECHNOLOGY LAB

(Electronics, Mechanical, Logic & Computational)

BASIC EQUIPMENT (Starter Level)

Item	Quantity	Unit Price (INR)	Total Price (INR)	Specifications
Electronics Lab				
Basic Breadboards & Components	20 sets	1,500	30,000	Resistors, capacitors, LEDs
Digital Multimeters	15 units	1,500	22,500	Basic measurement
DC Power Supplies	8 units	3,500	28,000	Variable voltage/current
Basic Oscilloscope	4 units	15,000	60,000	Entry-level, 2-channel
Soldering Stations	10 units	2,500	25,000	Temperature controlled
Component Storage Systems	5 systems	2,000	10,000	Organized component storage
Mechanical Lab				
3D Printer (Entry Level)	2 units	35,000	70,000	PLA printing, 200x200 bed
Basic Hand Tools Set	10 sets	1,500	15,000	Screwdrivers, wrenches, pliers

Gear & Pulley Demo Kits	6 sets	3,000	18,000	Mechanical principles
CAD Workstations	8 units	40,000	3,20,000	Design software capable
Logic & Programming				
Logic Simulation Software	1 license	15,000	15,000	Logisim, circuit simulation
Microcontroller Trainer Kits	15 sets	2,500	37,500	8051/Arduino based
FPGA Development Boards	8 units	5,000	40,000	Basic programmable logic

BASIC EQUIPMENT SUBTOTAL: ₹6,91,000

ADVANCED EQUIPMENT (Pro Level)

Item	Quantity	Unit Price (INR)	Total Price (INR)	Specifications
Advanced Electronics				
Professional Oscilloscopes	6 units	45,000	2,70,000	4-channel, 100MHz
Spectrum Analyzers	2 units	1,25,000	2,50,000	RF analysis
Logic Analyzers	4 units	35,000	1,40,000	Digital signal analysis
PCB Design & Fabrication Setup	1 system	1,50,000	1,50,000	Professional PCB creation
Advanced Mechanical				
Industrial 3D Printer	1 unit	2,50,000	2,50,000	Metal/ceramic capable
CNC Training Machine	1 unit	3,50,000	3,50,000	3-axis milling machine

CAD/CAM Workstations	8 units	75,000	6,00,000	High-end design stations
Precision Measuring Tools	1 set	50,000	50,000	Calipers, micrometers
Computational Systems				
FPGA Advanced Development	6 units	25,000	1,50,000	High-end programmable logic
Embedded Systems Lab	1 setup	1,00,000	1,00,000	ARM, DSP platforms
Real-time Systems Lab	1 setup	80,000	80,000	RTOS development

ADVANCED EQUIPMENT SUBTOTAL: ₹21,40,000

FURNITURE & INFRASTRUCTURE

Item	Quantity	Unit Price (INR)	Total Price (INR)	Specifications
Specialized Furniture				
Electronics Workbenches	8 units	15,000	1,20,000	Anti-static, tool storage
Mechanical Workshop Tables	6 units	18,000	1,08,000	Heavy-duty, vice mounts
CAD Design Stations	8 units	10,000	80,000	Adjustable, ergonomic
Equipment Storage Cabinets	8 units	8,000	64,000	Tool and component storage
Safety & Environment				
Fume Extraction System	2 units	25,000	50,000	Soldering smoke removal

Safety Equipment Kit	1 set	15,000	15,000	Goggles, gloves, first aid
Power & Infrastructure				
Laboratory Power Distribution	1 system	75,000	75,000	Multiple voltage rails
UPS System (Industrial Grade)	1 unit	65,000	65,000	Clean power for sensitive equipment

FURNITURE & INFRASTRUCTURE SUBTOTAL: ₹5,77,000

MIXED TECHNOLOGY LAB TOTAL:

- **Basic Equipment:** ₹6,91,000
- **Advanced Equipment:** ₹21,40,000
- **Furniture & Infrastructure:** ₹5,77,000
- **GRAND TOTAL:** ₹34,08,000

 OVERALL SUMMARY

Lab Type	Basic Equipment	Advanced Equipment	Furniture & Infrastructure	Lab Total
Robotics & AI Lab	₹6,49,600	₹23,39,000	₹5,47,000	₹35,35,600
Coding & Programming Lab	₹15,26,500	₹33,50,000	₹5,87,000	₹54,63,500
Mixed Technology Lab	₹6,91,000	₹21,40,000	₹5,77,000	₹34,08,000

TOTAL INVESTMENT REQUIRED: ₹1,24,07,100

IMPLEMENTATION RECOMMENDATIONS

Phase 1: Foundation (Basic Equipment)

- **Total Investment:** ₹28,67,100
- **Timeline:** 3-6 months
- **Focus:** Core functionality for grades 6-8

Phase 2: Enhancement (Advanced Equipment)

- **Total Investment:** ₹78,29,000
- **Timeline:** 6-12 months after Phase 1
- **Focus:** Advanced projects for grades 9-12

Phase 3: Infrastructure

- **Total Investment:** ₹17,11,000
- **Timeline:** Parallel with Phase 1
- **Focus:** Supporting infrastructure

ADDITIONAL CONSIDERATIONS

Annual Operating Costs:

- **Consumables & Maintenance:** ₹2,00,000
- **Software License Renewals:** ₹1,50,000
- **Equipment Upgrades:** ₹3,00,000
- **Total Annual Operating:** ₹6,50,000

Training & Support:

- **Teacher Training Programs:** ₹2,00,000
- **Technical Support Contract:** ₹1,50,000
- **Curriculum Development:** ₹1,00,000

Space Requirements:

- **Robotics & AI Lab:** 800-1000 sq ft

- **Coding Lab:** 600-800 sq ft
- **Mixed Technology Lab:** 1000-1200 sq ft
- **Total Space Needed:** 2400-3000 sq ft

*Note: Prices are approximate and based on current Indian market rates as of August 2025.
Actual costs may vary based on suppliers, bulk discounts, and specific requirements.
Consider phased implementation to manage budget effectively.*